

Computer Vision, Lecture 3

Darius Burschka

<http://www6.in.tum.de/burschka/courses/vision/>

Outline for Today

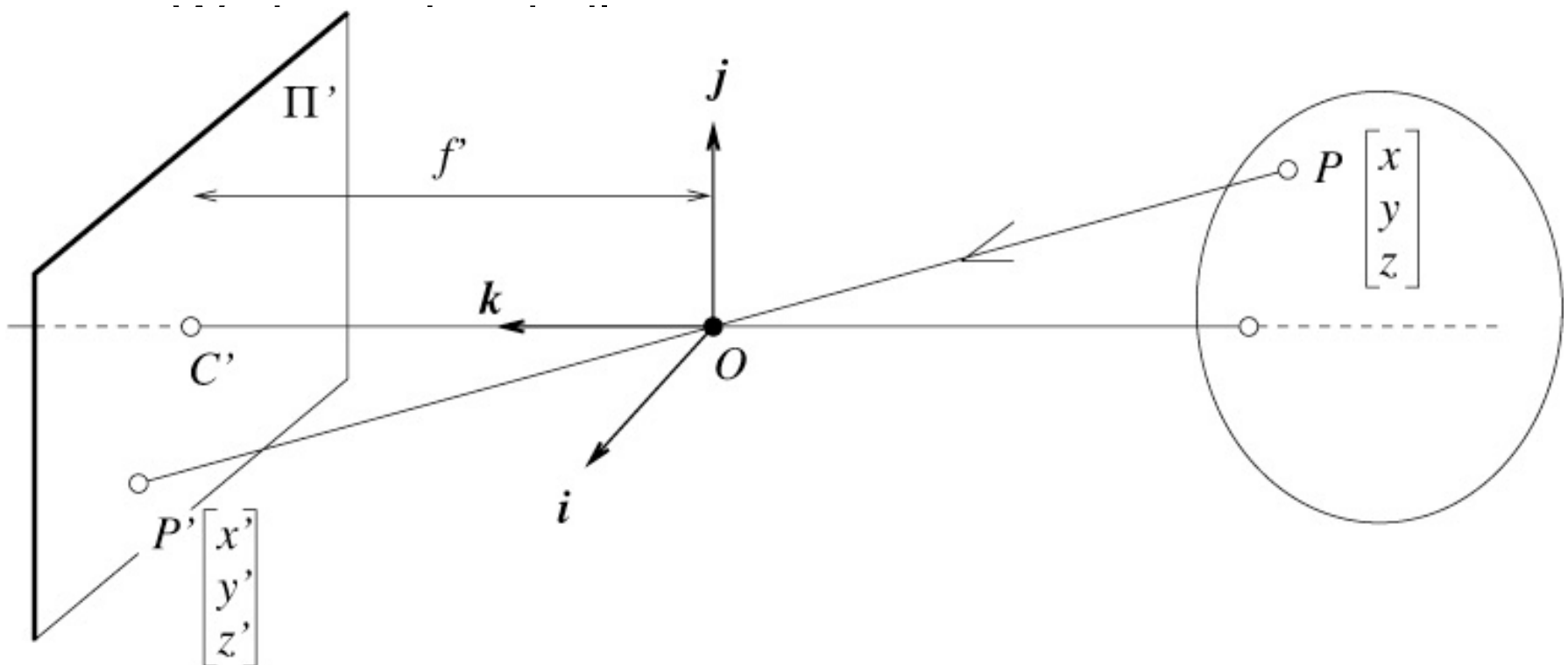
- Review of Camera Models
 - Continue to discuss geometric transformations
- Some special properties of Camera Models
- Reflectance



The equation of projection

$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

- Cartesian coordinates:

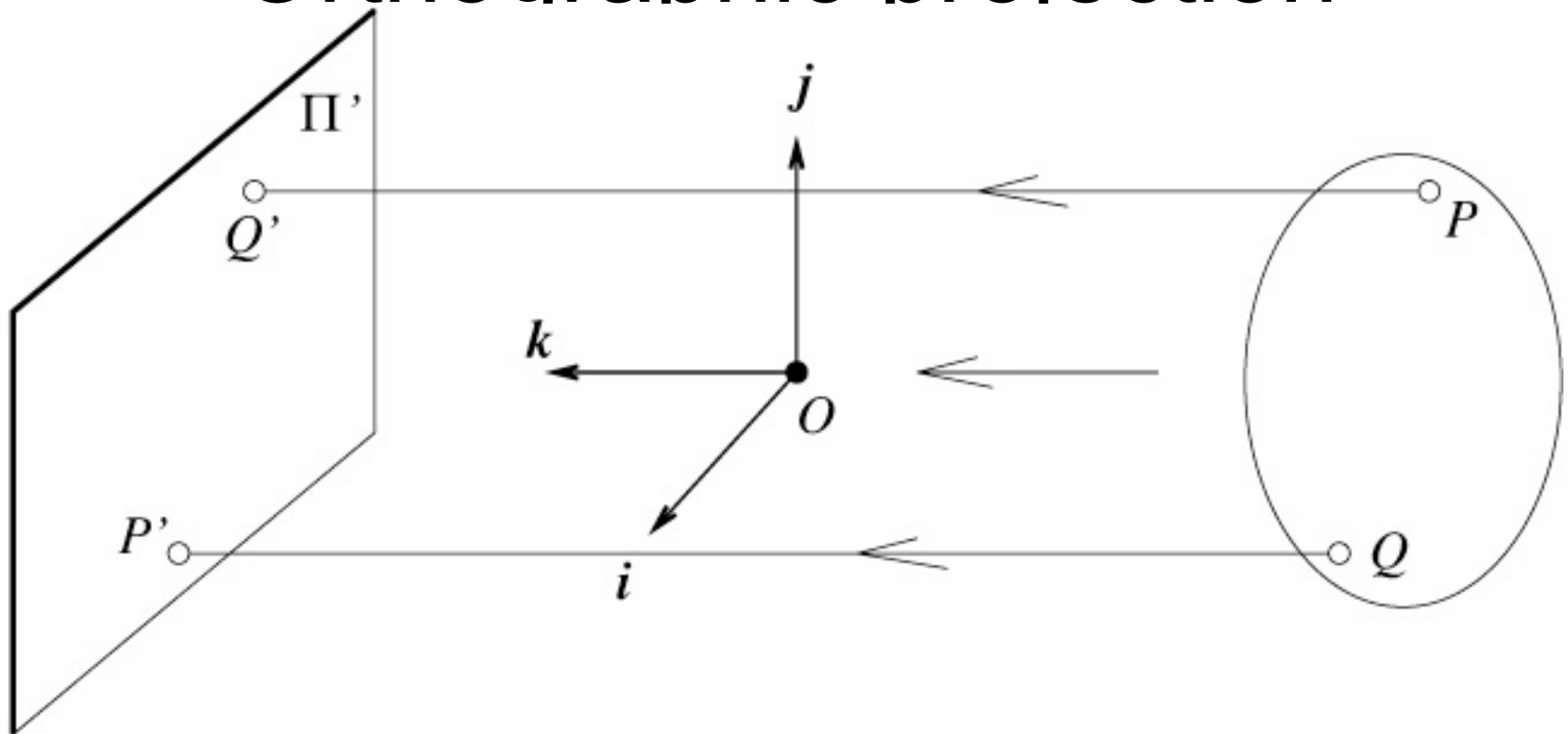


The camera matrix

- Homogenous coordinates for 3D
 - four coordinates for 3D point
 - equivalence relation (X,Y,Z,T) is the same as (kX, kY, kZ, kT)
- Turn previous expression into HC's
 - HC's for 3D point are (X,Y,Z,T)
 - HC's for point in image are (U,V,W)

$$\begin{pmatrix} U \\ V \\ W \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1/f & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ T \end{pmatrix} \quad (U,V,W) \rightarrow \left(\frac{U}{W}, \frac{V}{W}\right) = (u,v)$$

Orthographic projection



$$u = x$$

$$v = y$$

The fundamental model for orthographic projection

$$\begin{pmatrix} U \\ V \\ W \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ T \end{pmatrix}$$

Weak perspective

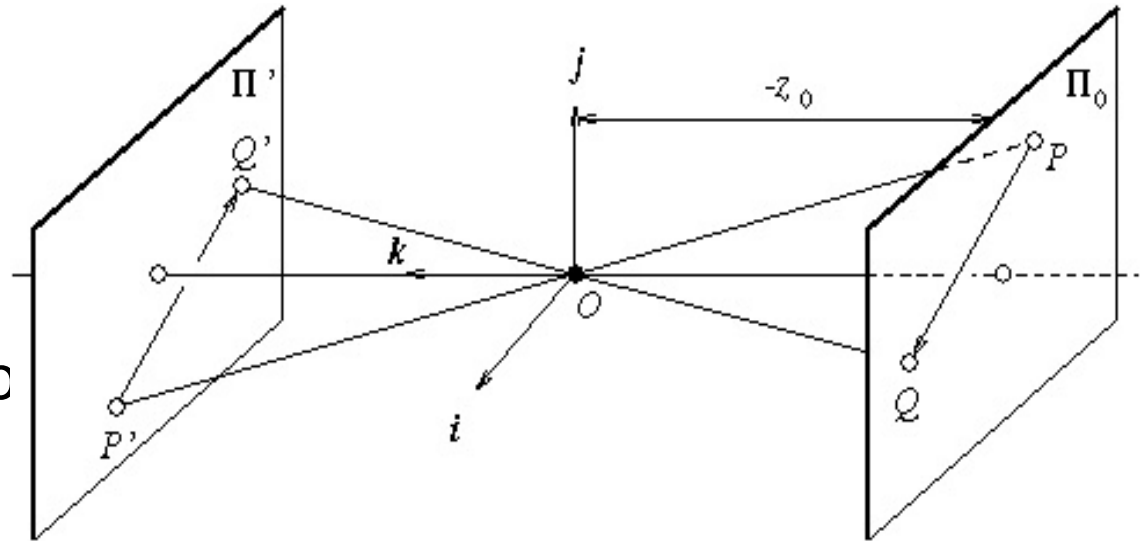
$$u = Tx$$

$$v = Ty$$

$$T = f / Z *$$

- Issue

- perspective effects, but not over the scale of individual objects
- collect points into a group at about the same depth, then divide each point by the depth of its group
- Adv: easy
- Disadv: wrong



The fundamental model for orthographic projection

$$\begin{pmatrix} U \\ V \\ W \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & f / Z^* \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ T \end{pmatrix}$$

Camera parameters

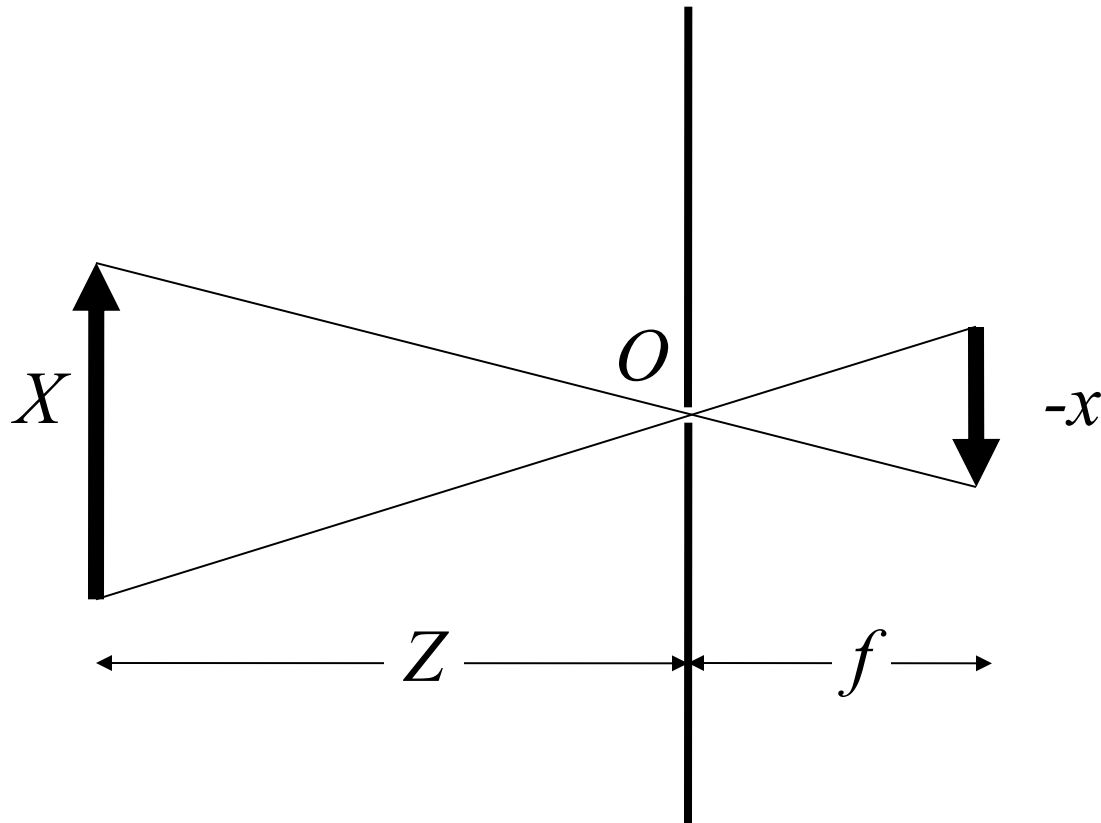
- Issue
 - camera may not be at the origin, looking down the z-axis
 - extrinsic parameters
 - one unit in camera coordinates may not be the same as one unit in world coordinates
 - intrinsic parameters - focal length, principal point, aspect ratio, angle between axes, etc.

$$\begin{pmatrix} U \\ V \\ W \end{pmatrix} = \begin{pmatrix} \text{Transformation} \\ \text{representing} \\ \text{intrinsic parameters} \end{pmatrix} \begin{pmatrix} \text{Transformation} \\ \text{representing} \\ \text{projection model} \end{pmatrix} \begin{pmatrix} \text{Transformation} \\ \text{representing} \\ \text{extrinsic parameters} \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ T \end{pmatrix}$$

Intrinsic Camera Parameters

- Determine the intrinsic parameters of a camera (with lens)
- What are *Intrinsic Parameters*?

Perspective Projection, Remember?

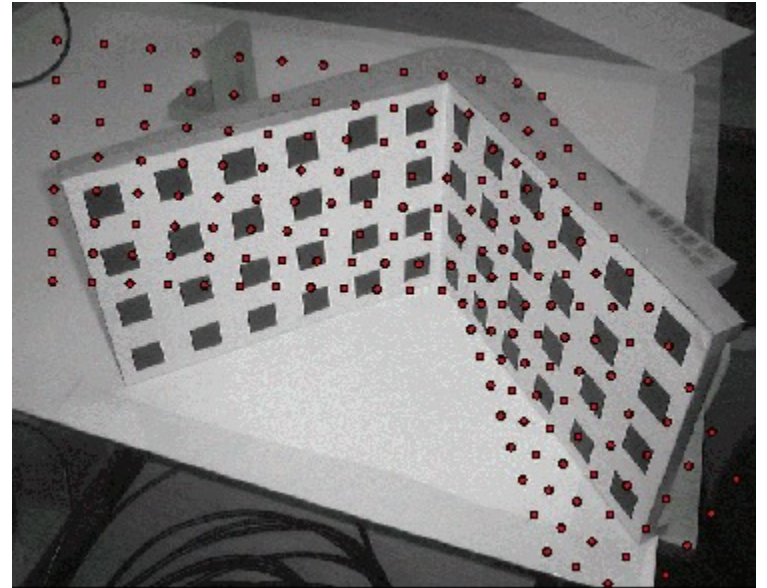
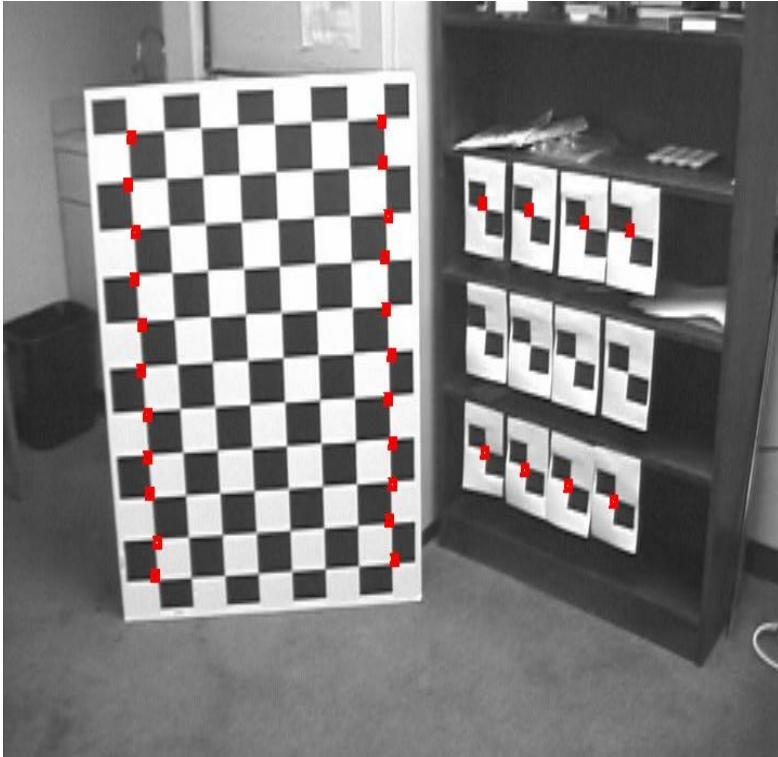


$$x = X \frac{f}{Z}$$

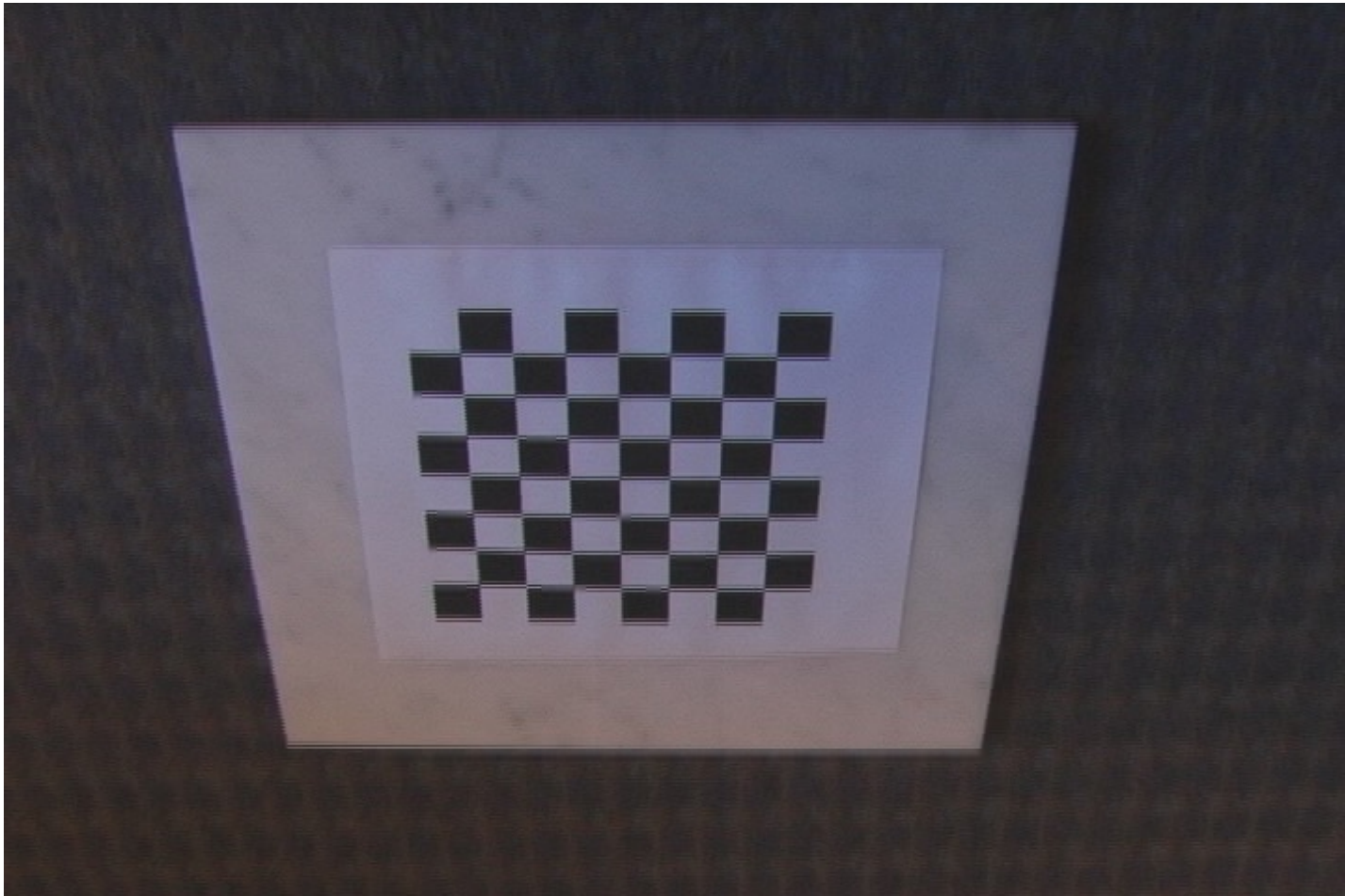
Intrinsic Camera Parameters

- Determine the intrinsic parameters of a camera (with lens)
- Intrinsic *Parameters*:
 - Focal Length f
 - Pixel size s_x, s_y
 - Distortion coefficients $k_1, k_2 \dots$
 - Image center o_x, o_y

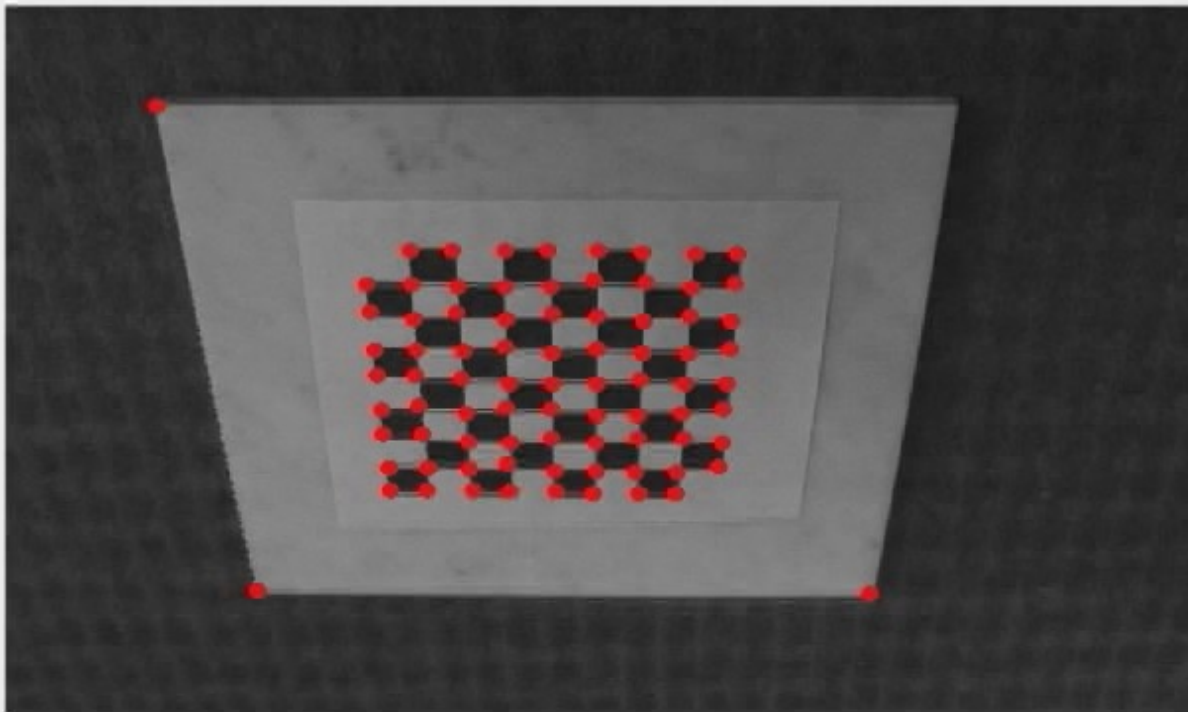
Example Calibration Pattern



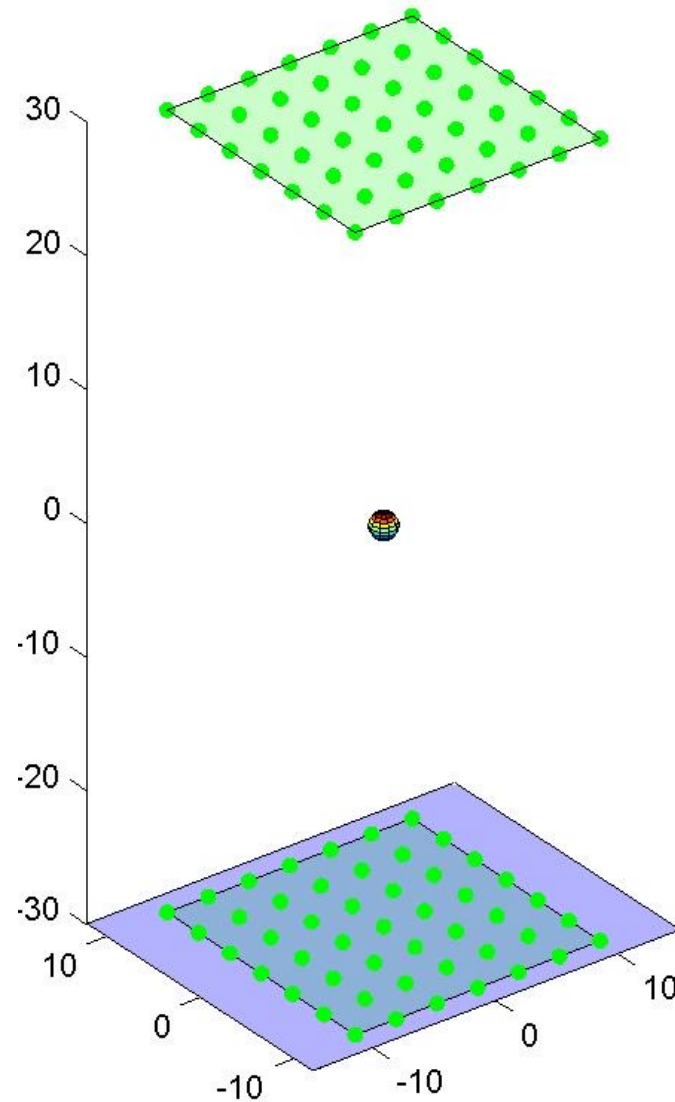
Our Calibration target



Harris Corner Detector



Experiment 1: Parallel Board

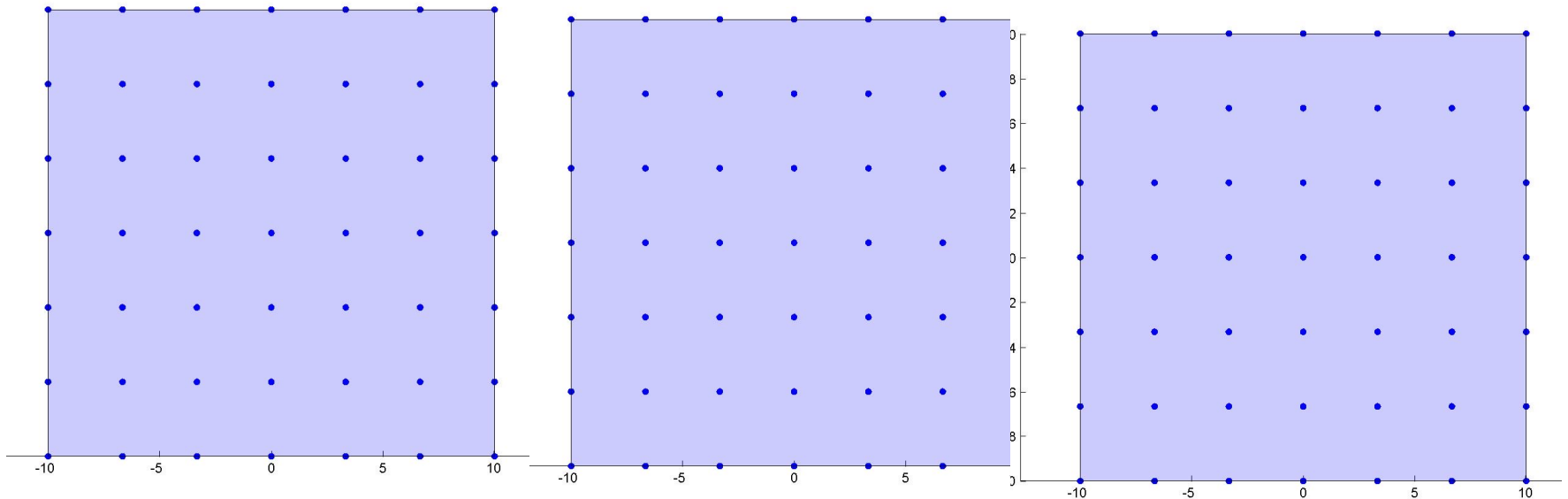


Projective Perspective of Parallel Board

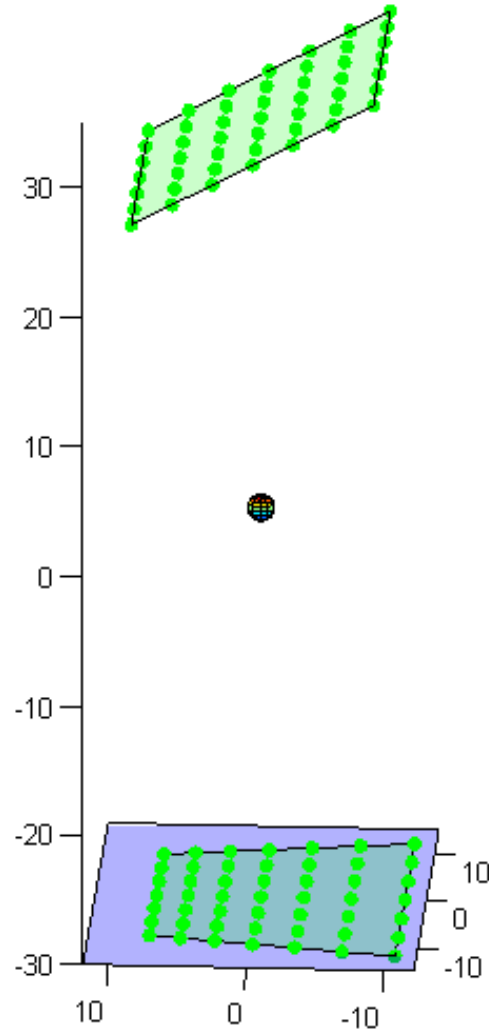
10cm

20cm

30cm

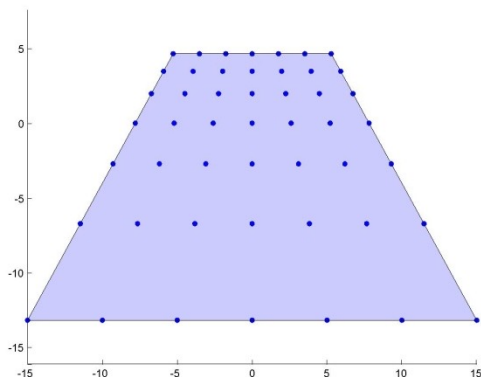


Experiment 2: Tilted Board

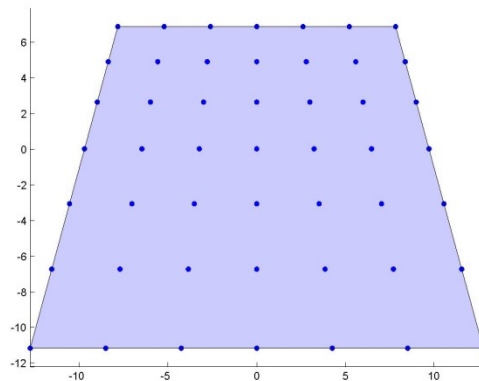


Projective Perspective of Tilted Board

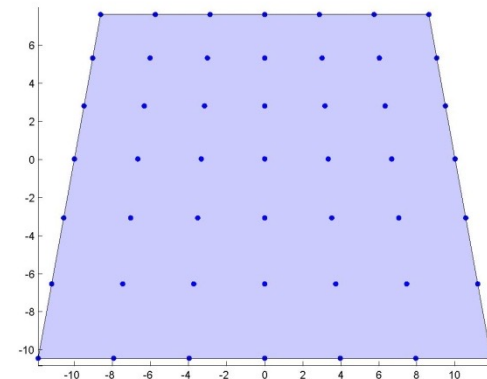
10cm



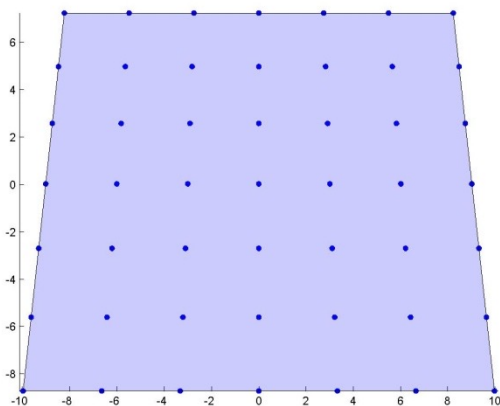
20cm



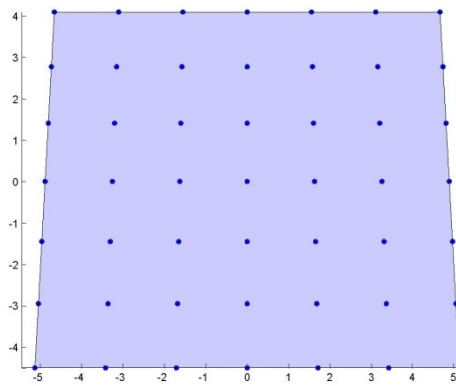
30cm



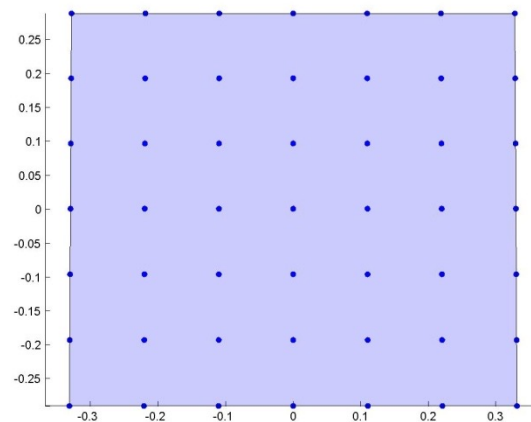
50cm



100cm

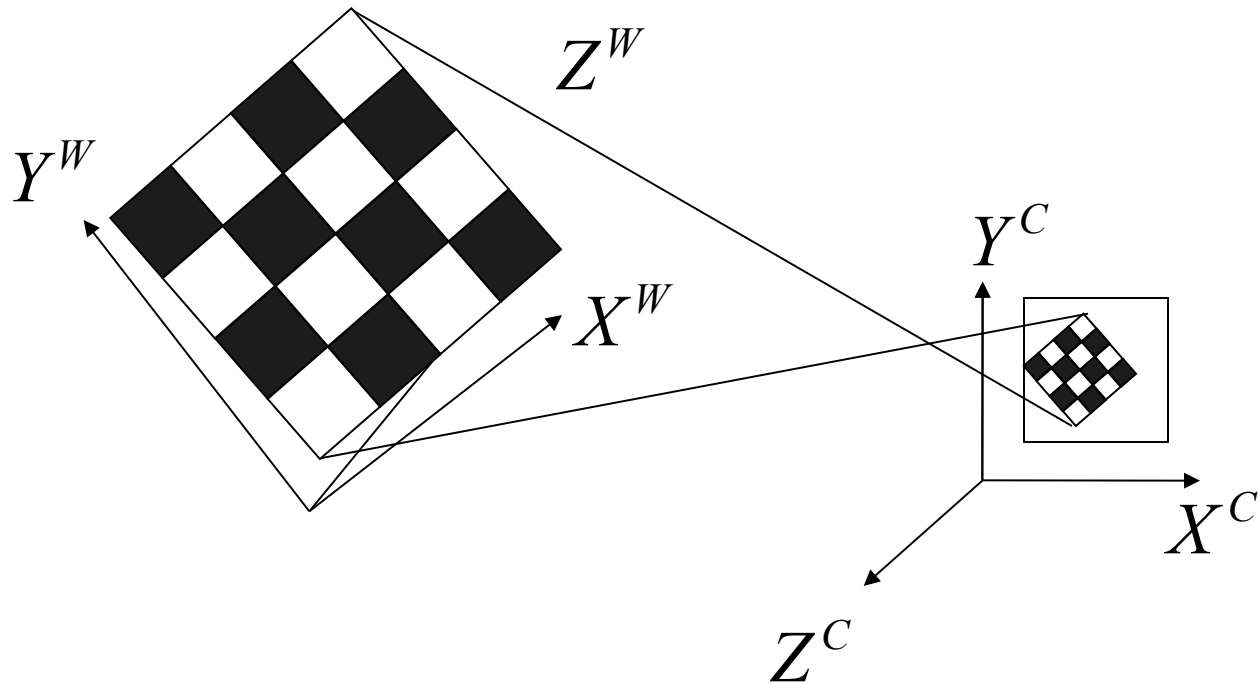


500cm



Perspective Camera Model

Object Space



Calibration: 2 steps

- Step 1: Transform into camera coordinates

$$\begin{pmatrix} \tilde{X}^c \\ \tilde{Y}^c \\ \tilde{Z}^c \end{pmatrix} = f \left(\begin{pmatrix} X^W \\ Y^W \\ Z^W \end{pmatrix}, \phi, \varphi, \psi, T \right)$$
$$x_{im} = -\frac{f}{s_x} \frac{\tilde{X}^c}{\tilde{Z}^c} + o_x$$
$$y_{im} = -\frac{f}{s_y} \frac{\tilde{Y}^c}{\tilde{Z}^c} + o_y$$

- Step 2: Transform into image coordinates

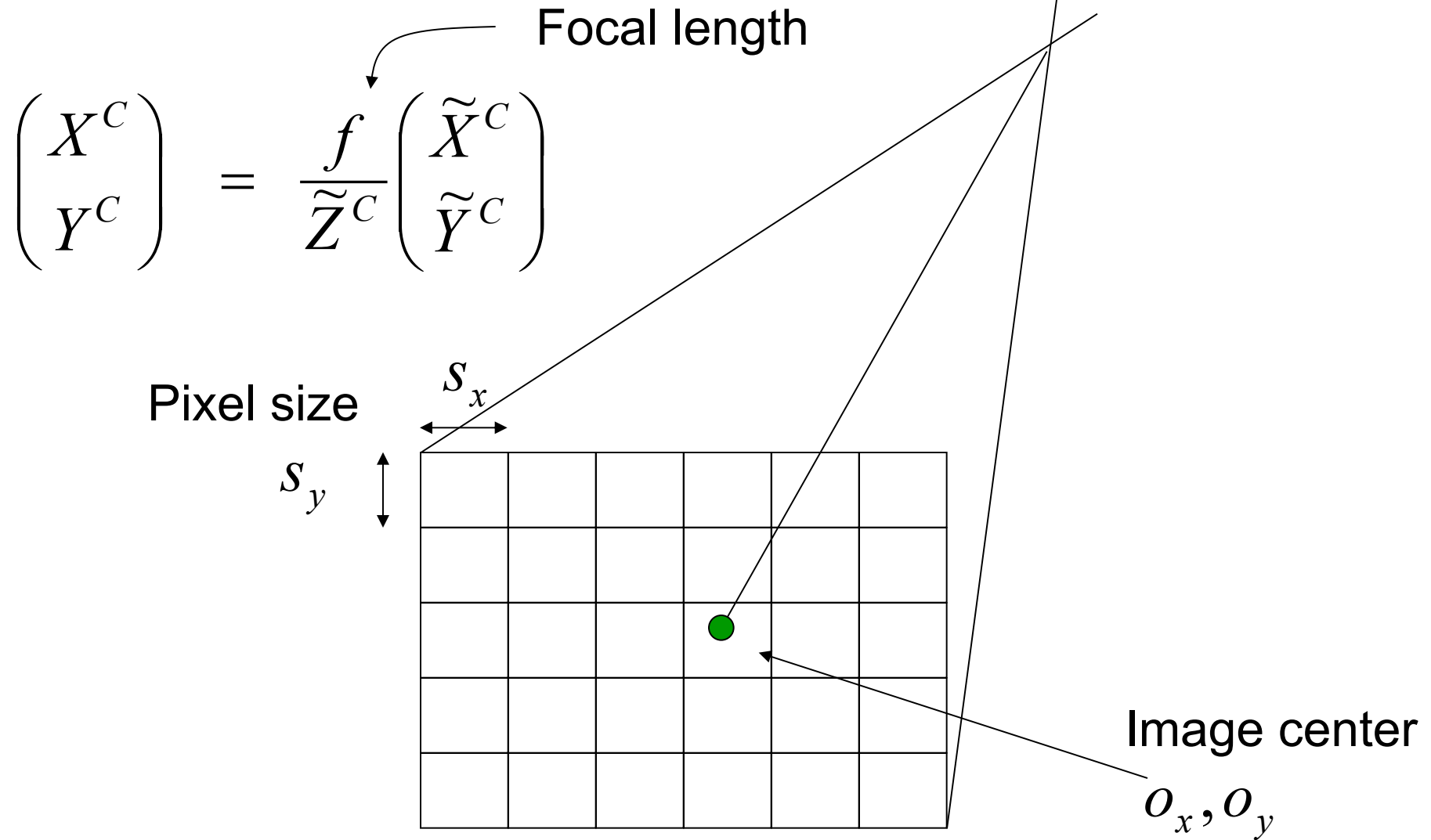
Calibration Model (extrinsic)

Homogeneous Coordinates

$$\begin{pmatrix} \tilde{X}^c \\ \tilde{Y}^c \\ \tilde{Z}^c \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \varphi & 0 & \sin \varphi \\ & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & \sin \psi \\ 0 & -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} X^w \\ Y^w \\ Z^w \end{pmatrix} + \begin{pmatrix} T_x \\ T_y \\ T_z \end{pmatrix}$$

$$\begin{pmatrix} X^c \\ Y^c \end{pmatrix} = \frac{f}{\tilde{Z}^c} \begin{pmatrix} \tilde{X}^c \\ \tilde{Y}^c \end{pmatrix}$$

Calibration Model (intrinsic)



Intrinsic Transformation

$$\begin{pmatrix} x_{im} \\ y_{im} \end{pmatrix} = \begin{bmatrix} \left(-\frac{f}{s_x} \frac{\tilde{X}^c}{\tilde{Z}^c} + o_x + .5 \right) \\ \left(-\frac{f}{s_y} \frac{\tilde{Y}^c}{\tilde{Z}^c} + o_y + .5 \right) \end{bmatrix}$$

Plugging the Model Together!

$$\begin{pmatrix} x_{im} \\ y_{im} \end{pmatrix} = \begin{pmatrix} -\frac{f}{s_x} \frac{(\cos \phi \quad \sin \phi \quad 0) \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & \sin \psi \\ 0 & -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} X^W[i] \\ Y^W[i] \\ Z^W[i] \end{pmatrix} + T_X}{(0 \quad 0 \quad 1) \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & \sin \psi \\ 0 & -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} X^W[i] \\ Y^W[i] \\ Z^W[i] \end{pmatrix} + T_z} + o_x + .5 \\ -\frac{f}{s_y} \frac{(-\sin \phi \quad \cos \phi \quad 0) \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & \sin \psi \\ 0 & -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} X^W[i] \\ Y^W[i] \\ Z^W[i] \end{pmatrix} + T_Y}{(0 \quad 0 \quad 1) \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & \sin \psi \\ 0 & -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} X^W[i] \\ Y^W[i] \\ Z^W[i] \end{pmatrix} + T_z} + o_y + .5 \end{pmatrix}$$

Summary Parameters

- Extrinsic

- Rotation ϕ, φ, ψ
- Translation T

- Intrinsic

- Focal length
- Pixel size
- Image center coordinates
- (Distortion coefficients)

$$f, (s_x, s_y), (o_x, o_y), k_1, \dots$$

Q: Can We recover all Intrinsic Params?

- No

Summary Parameters, Revisited

- Extrinsic

- Rotation ϕ, φ, ψ

- Translation T

Focal length, in pixel units $\frac{f}{s_x}$

Aspect ratio $\alpha = \frac{s_x}{s_y}$

- Intrinsic

- ~~– Focal length f~~

- ~~– Pixel size (s_x, s_y)~~

- Image center coordinates

- (Distortion coefficients)

$k_1, \dots$

Calibration via SVD

$$x - \phi_x = -f_x \frac{r_{11}X^W + r_{12}Y^W + r_{13}Z^W + T_x}{r_{31}X^W + r_{32}Y^W + r_{33}Z^W + T_z}$$
$$y - \phi_y = -f_y \frac{r_{21}X^W + r_{22}Y^W + r_{23}Z^W + T_y}{r_{31}X^W + r_{32}Y^W + r_{33}Z^W + T_z}$$

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} \quad T = \begin{pmatrix} T_x \\ T_y \\ T_z \end{pmatrix}$$

Calibration via SVD

$$x_i = -f_x \frac{r_{11}X_i^W + r_{12}Y_i^W + r_{13}Z_i^W + T_x}{r_{31}X_i^W + r_{32}Y_i^W + r_{33}Z_i^W + T_z}$$

$$y_i = -f_y \frac{r_{21}X_i^W + r_{22}Y_i^W + r_{23}Z_i^W + T_y}{r_{31}X_i^W + r_{32}Y_i^W + r_{33}Z_i^W + T_z}$$

$$x_i f_y (r_{21}X_i^W + r_{22}Y_i^W + r_{23}Z_i^W + T_y) = y_i f_x (r_{11}X_i^W + r_{12}Y_i^W + r_{13}Z_i^W + T_x)$$

N>=7 points, not coplanar
31

Calibration via SVD

$$x_i f_y (r_{21} X_i^W + r_{22} Y_i^W + r_{23} Z_i^W + T_y) = y_i f_x (r_{11} X_i^W + r_{12} Y_i^W + r_{13} Z_i^W + T_x)$$

$$x_i X_i^W v_1 + x_i Y_i^W v_2 + x_i Z_i^W v_3 + x_i v_4 - y_i X_i^W v_5 - y_i Y_i^W v_6 - y_i Z_i^W v_7 - y_i v_8 = 0$$

$$\alpha = \frac{f_x}{f_y}$$

$$v_1 = r_{21} \quad v_5 = \alpha r_{11}$$

$$v_2 = r_{22} \quad v_6 = \alpha r_{12}$$

$$v_3 = r_{23} \quad v_7 = \alpha r_{13}$$

$$v_4 = T_y \quad v_8 = \alpha T_x$$

Calibration via SVD

$$x_i X_i^W v_1 + x_i Y_i^W v_2 + x_i Z_i^W v_3 + x_i v_4 - y_i X_i^W v_5 - y_i Y_i^W v_6 - y_i Z_i^W v_7 - y_i v_8 = 0$$

$$A = \begin{bmatrix} x_1 X_1^W & x_1 Y_1^W & x_1 Z_1^W & x_1 & -y_1 X_1^W & -y_1 Y_1^W & -y_1 Z_1^W & -y_1 \\ x_2 X_2^W & x_2 Y_2^W & x_2 Z_2^W & x_2 & -y_2 X_2^W & -y_2 Y_2^W & -y_2 Z_2^W & -y_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_N X_N^W & x_N Y_N^W & x_N Z_N^W & x_N & -y_N X_N^W & -y_N Y_N^W & -y_N Z_N^W & -y_N \end{bmatrix}$$

$$Av = 0$$

A has rank 7 (without proof)

$$A = UDV^T \quad \text{via SVD (Singular Value Decomposition)}$$

Calibration via SVD

- Remaining Problem:

V doesn't give a rotation matrix

$$v_1 = r_{21} \quad v_5 = \alpha r_{11}$$

$$v_2 = r_{22} \quad v_6 = \alpha r_{12}$$

$$v_3 = r_{23} \quad v_7 = \alpha r_{13}$$

$$v_4 = T_y \quad v_8 = \alpha T_x$$

- See book

Summary, SVD Solution

- Replace rotation matrix by arbitrary matrix
- Transform into linear set of equations
- Solve via SVD
- Enforce rotation matrix (see book)
- Solve for remaining parameters (see book)

SVD solution: algebraic minimization, assume Gaussian noise in parameter space

Calibration by nonlinear Least Squares

- Calibration Examples:

$$\begin{pmatrix} x_{im}[1] \\ y_{im}[1] \end{pmatrix} = g\left(\begin{pmatrix} X^W[1] \\ Y^W[1] \\ Z^W[1] \end{pmatrix}, \phi, \varphi, \psi, T, \frac{f}{s_x} \alpha, o_x, o_y\right)$$

$$\begin{pmatrix} x_{im}[2] \\ y_{im}[2] \end{pmatrix} = g\left(\begin{pmatrix} X^W[2] \\ Y^W[2] \\ Z^W[2] \end{pmatrix}, \phi, \varphi, \psi, T, \frac{f}{s_x} \alpha, o_x, o_y\right)$$

•
•
•

$$\begin{pmatrix} x_{im}[N] \\ y_{im}[N] \end{pmatrix} = g\left(\begin{pmatrix} X^W[N] \\ Y^W[N] \\ Z^W[N] \end{pmatrix}, \phi, \varphi, \psi, T, \frac{f}{s_x} \alpha, o_x, o_y\right)$$

Calibration by nonlinear Least Squares

- Least Squares

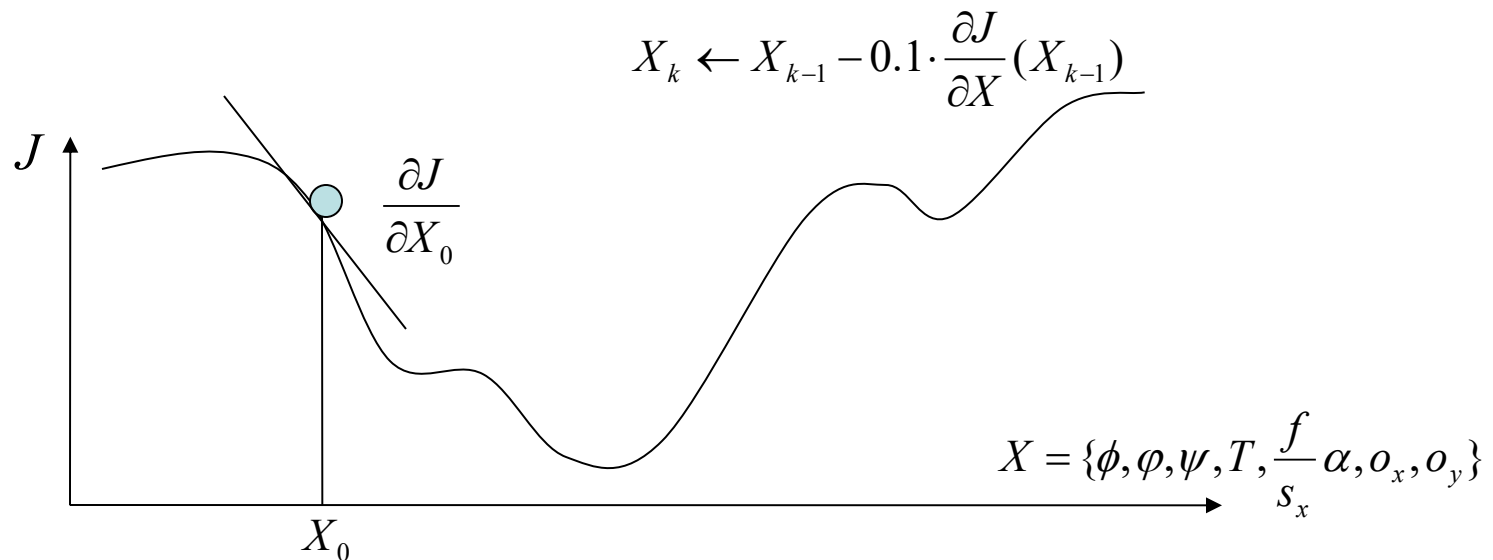
$$J = \sum_i \left[\begin{pmatrix} x_{im}[i] \\ y_{im}[i] \end{pmatrix} - g \left(\begin{pmatrix} X^W[i] \\ Y^W[i] \\ Z^W[i] \end{pmatrix}, \phi, \varphi, \psi, T, \frac{f}{s_x} \alpha, o_x, o_y \right) \right]^2 \rightarrow \min$$

Calibration by nonlinear Least Squares

- Least Mean Square

$$J = \sum_i \left[\begin{pmatrix} x_{im}[i] \\ y_{im}[i] \end{pmatrix} - g \left(\begin{pmatrix} X^W[i] \\ Y^W[i] \\ Z^W[i] \end{pmatrix}, \phi, \varphi, \psi, T, \frac{f}{s_x} \alpha, o_x, o_y \right) \right]^2 \rightarrow \min$$

- Gradient descent:



Summary Non-Linear Least Squares

- Solve nonlinear equations via gradient descent
- Assume Gaussian noise in image space, not parameter space

SVD Versus LQ

SVD

- Minimization of squared distance in parameter space
- Globally optimal

Nonlin Least Squares

- Minimization of squared distance in Image space
- Locally optimal

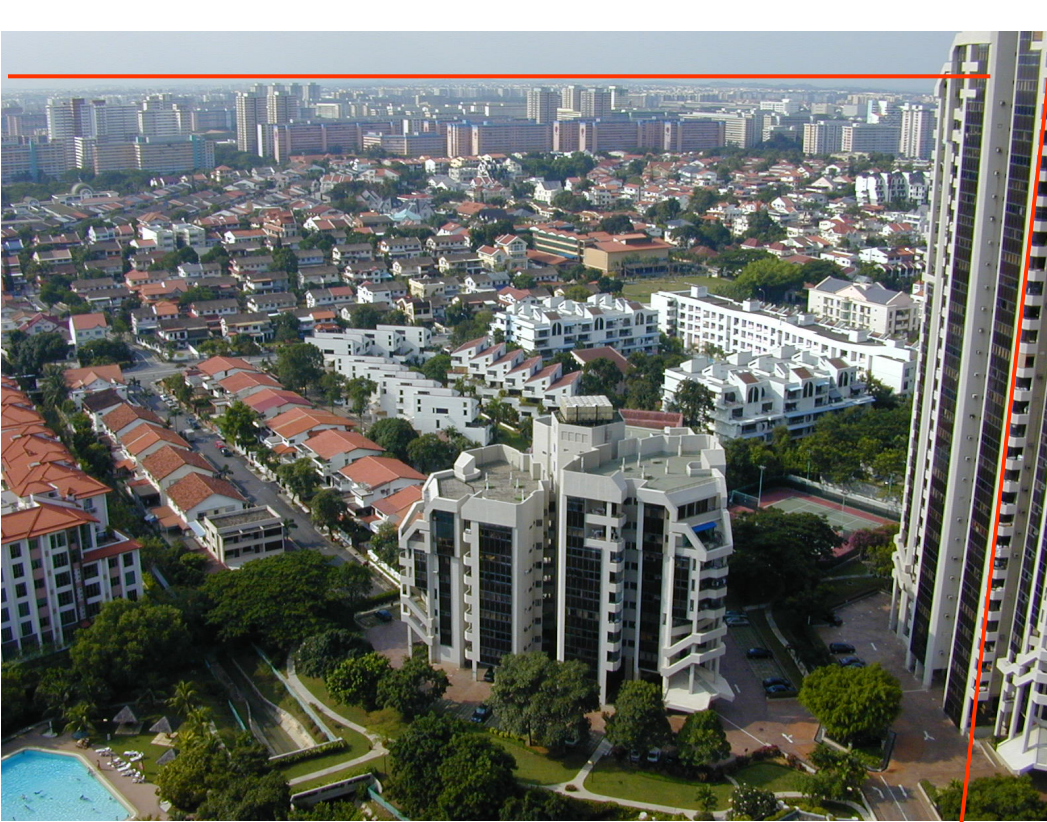
Q: How Many Images Do We Need?

- Assumption: K images with M corners each
- $4+6K$ parameters
- $2KM$ constraints
- $2KM \geq 4+6K \Rightarrow M > 3$ and $K \geq 2/(M-3)$
- 2 images with 4 points, but will 1 images with 5 points work?
- No, since points cannot be co-planar!

Advanced Calibration: Nonlinear Distortions

- Barrel and Pincushion
- Tangential

Barrel and Pincushion Distortion



wideangle



tele

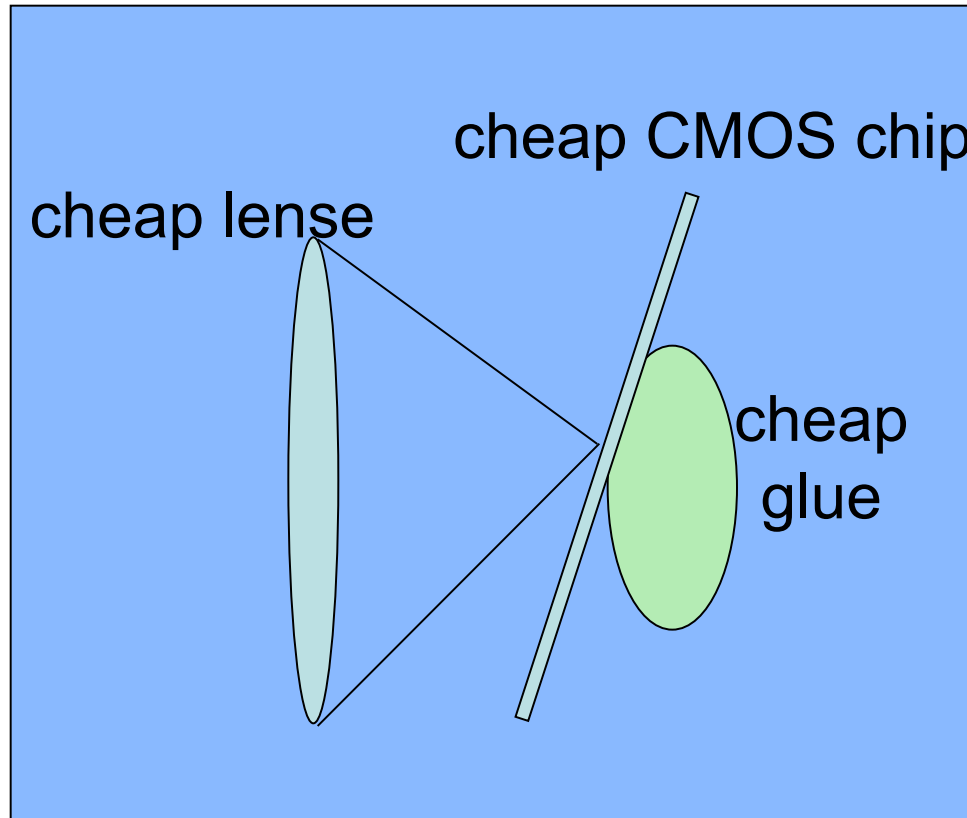
Models of Radial Distortion

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_d \\ y_d \end{pmatrix} (1 + k_1 r^2 + k_2 r^4)$$

distance from center



Tangential Distortion



cheap camera

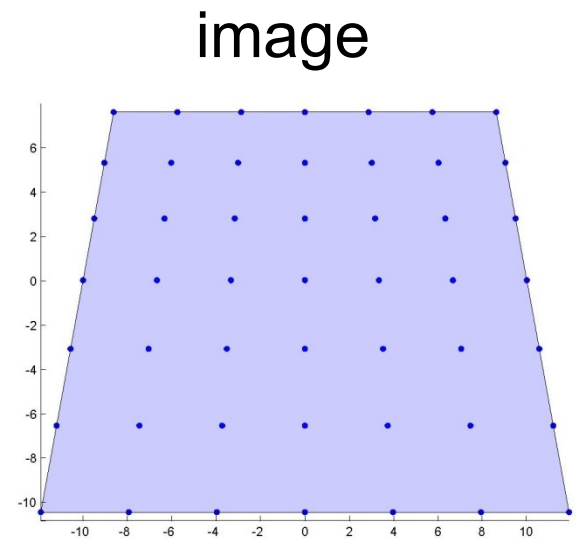
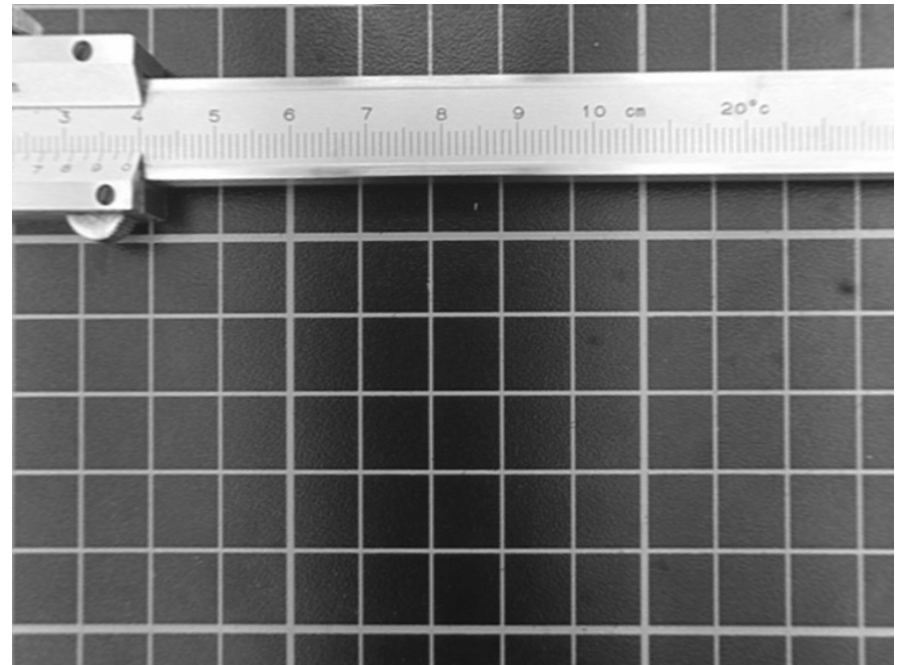
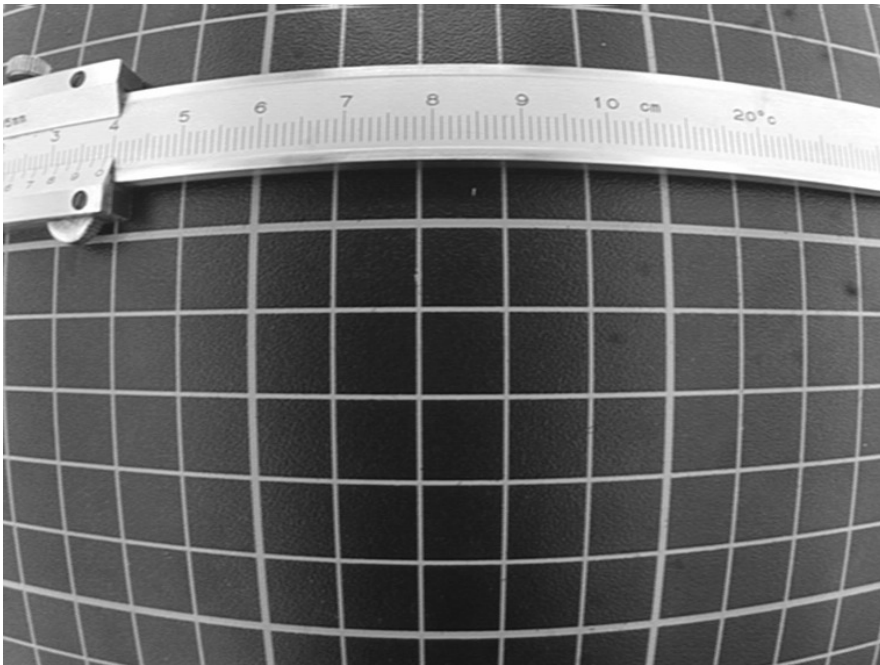


Image Rectification (to be continued)



Summary

- Calibration: Problem definition
- Solution via Singular Value Decomposition
- Solution by nonlinear Least Squares
- Distortion

Intrinsic Parameters

Intrinsic Parameters describe the conversion from metric to pixel coordinates (and the reverse)

$$x_{mm} = - (x_{pix} - o_x) s_x$$

$$y_{mm} = - (y_{pix} - o_y) s_y$$

or

$$\begin{pmatrix} x \\ y \\ w \end{pmatrix}_{pix} = \begin{pmatrix} -f / s_x & 0 & o_x \\ 0 & -f / s_y & o_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ w \end{pmatrix}_{mm} = M_{int} p$$

Camera Model Structure

Assume R and T express camera *in world coordinates*, then

$${}^c p = \begin{pmatrix} R' & -R' T \\ 0 & 0 & 0 & 1 \end{pmatrix}^w p$$

Combining with a perspective model (and neglecting internal parameters) yields

$${}^c u = M^w p = \begin{pmatrix} -R'_x & R'_x T \\ -R'_y & R'_y T \\ \frac{R'_z}{f} & \frac{-R'_z T}{f} \end{pmatrix}^w p$$

Note the M is defined only up to a scale factor at this point! If M is viewed as a 3×4 matrix defined up to scale, it is called the *projection matrix*.

Camera Model Structure

Assume R and T express camera *in world coordinates*, then

$${}^c p = \begin{pmatrix} R' & -R' T \\ 0 & 0 & 0 & 1 \end{pmatrix}^w p$$

Combining with a weak perspective model (and neglecting internal parameters) yields

$${}^c u = M^w p = \begin{pmatrix} -R'_x & R'_x T \\ -R'_y & R'_y T \\ 0 & \frac{R'_z(\bar{P} - T)}{f} \end{pmatrix}^w p$$

Where $\bar{P}$ is the nominal distance to the viewed object

Lens Distortion

- In general, lens introduce minor irregularities into images, typically radial distortions:

$$x = x_d(1 + k_1r^2 + k_2r^4)$$

$$y = y_d(1 + k_1r^2 + k_2r^4)$$

$$r^2 = x_d^2 + y_d^2$$

- The values k_1 and k_2 are additional parameters that must be estimated in order to have a model for the camera system.